

SHAUNAK RANE

Dombivli,Thane,Maharashtra - shaunak.rane@universalai.in

SUMMARY

Innovative AI & Software Engineer with strong expertise in Python, Machine Learning, and Backend Architecture. Proven track record of building scalable systems, from Autonomous Multi-Agent AI to Microservices-based web applications. Proficient in Cloud Infrastructure, API Design (FastAPI), and Database Optimization. Passionate about solving complex problems by combining data science with robust software engineering principles.

WORK EXPERIENCE

Univitt Ai Technologies

April 2025 - July 2025

Software Project Trainee

Architected a high-throughput, asynchronous ASGI backend utilizing Python and FastAPI for real-time state management and event-driven expiration telemetry, mitigating asset spoilage by an estimated 25%. The underlying infrastructure leverages a fully normalized, ACID-compliant SQL database, optimized via advanced query execution mapping to handle distributed donation logistics and dynamic lifecycle routing. For predictive resource allocation, I trained and deployed a Random Forest Regression ensemble model directly into the inference pipeline, executing complex time-series demand forecasting to drive automated procurement and maximize systemic stock utilization.

PROJECTS

Project Aegis | Agentic AI - Misinformation

- Architected a real-time, multi-agent orchestration system using FastAPI and the Gemini API to autonomously profile sources, scrape web evidence, and debunk misinformation.
- Engineered a custom ML classification model and a high-throughput data pipeline via Supabase (PostgreSQL) to stream live threat telemetry to a dashboard using WebSockets.
- Designed and deployed robust RESTful API endpoints for system observability, agent state debugging, and mock data injection to ensure fault tolerance and maintainability.
- Live : <https://agentai100.netlify.app>

Yield AI | KrushiMitra – Farmers Friend

- Developed a mobile-first Single Page Application (SPA) using React and TypeScript, utilizing Radix UI and Tailwind CSS for an accessible, component-driven interface.
- Implemented a decoupled microservices architecture, isolating core routing logic from compute-heavy prediction services to optimize asynchronous data pipelines and scalability
- Integrated dynamic data visualization to translate complex ML inference payloads (yield probability, confidence intervals) into actionable, user-friendly metrics for agricultural decision-making.
- Github: <https://github.com/Shounakrane914/Yield-Ai>

Gridium Protocol (AI-Optimized Blockchain Energy Market)

- Architected a decentralized, blockchain-based Local Energy Market (LEM) combining an Ethereum smart contract settlement layer with a Python FastAPI backend and React Three Fiber 3D frontend.
 - Engineered a Deep Deterministic Policy Gradient (DDPG) reinforcement learning agent to autonomously balance physical microgrid loads by dynamically adjusting Automated Market Maker (AMM) trading fees.
 - Deployed a constant-product AMM via Ethereum smart contracts, integrating Groth16 zero-knowledge proofs (zk-SNARKs) to cryptographically verify peer-to-peer energy trades while guaranteeing 100% user data privacy..
 - Live : <https://gridiumprotocol.netlify.app>
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SKILLS

- Technical Skills: Python, JavaScript, C, FastAPI, REST APIs,Pandas, NumPy, Scikit-learn, Data Preprocessing, Data Visualization,MySQL, NoSQL, Generative AI, RAG Fundamentals, Problem Solving
- Certifications: — Generative Ai, Learning C Programming,Ai for everyone, Introduction to LLM,etc